# Cost-first $\kappa_3$ architecture assessment

A negative result: no qualifying representation established
10 October 2026

**[D/S] I do not have a defensible new representation that meets both 116.4 u and the stated 2% channel tolerance.** I would not repackage a measured-dead compression or slice recursion as a solution. The useful conclusions below are a sharper budget requirement, an exact missing-information identity, and a cheap test for future claims. They are not an architecture claimed to pass.

**Labels:** [D] derived; [E] implementation estimate; [S] speculative or unestablished; [Q] quoted from your prompt. No experiments or network measurements were run. I interpret 2% [Q] as replacement-versus-current-chain error. If it means error versus the teacher, this also asks for an accuracy improvement over your previously reported chain errors.

## The dominant work must fall by more than half

[D from Q] Use 194 u as the baseline. The approximately 70 u of slice readouts and 63 u of young transport total 133 u. Keep the remaining 61 u as a bookkeeping allowance, including rounding of the ledger. Then:

```
target = 0.60 * 194 = 116.4 u
remaining hot-work budget = 116.4 - 61 = 55.4 u
required hot-work reduction = 1 - 55.4/133 = 58.35%
```

| Hypothetical saving [D] | Remaining total |
|---|---|
| Eliminate all 70 u of slice readout | 124 u |
| Eliminate all 63 u of young transport | 131 u |
| Halve both families | 127.5 u |
| Requested 60% of baseline | 116.4 u |

[D] These are ledger bounds with all other work fixed, not universal algorithmic lower bounds. A viable design must also reduce part of the remainder, or remove more than half of the dominant work. With about 70 young source-layers [Q], the hot-work allowance is about 0.791 u per source-layer [D], compared with the quoted 1.9 u [Q]. The exact score floor is 102.4 u [D].

## What the leaderboard does and does not establish

[D] Matching final mean MSE does not imply matching your D21 or d3. A different estimator may have different, jointly compensating channel errors. The leaderboard therefore demonstrates that a cheaper estimator exists; it does not demonstrate a cheaper drop-in carrier preserving this chain's cumulant inputs to 2%.

[S] The leaders may use a different joint closure or a representation outside the tested families, but I cannot infer which from their cost and final MSE. Your cancellation measurements [Q] are a reason not to invent a channel-equivalence claim. I cannot substantiate either “the leaders must drop sources” or “the floor is two products per source.”

# The precise obstruction to a slice state

This is an information argument, not a new proposal or a FLOP lower bound

[D] Let $S(T)_{ik}=T_{iik'}$ and let $U=W^{T}\mathrm{diag}(w_1)$. Choose distinct input indices a,b,c and an invisible symmetric cubic perturbation, with Sym denoting the average over permutations:

```
H = epsilon Sym(e_a, e_b, e_c)
S(H) = 0;   diag3(H) = 0
```

[D] Write $u_a,u_b,u_c$ for the corresponding columns of U. Its next slice and diagonal are:

```
S(U^tensor3 H) = (epsilon/3) * [
 (u_a*u_b) u_c.T + (u_a*u_c) u_b.T
                         + (u_b*u_c) u_a.T ]
diag3(U^tensor3 H) = epsilon u_a*u_b*u_c
```

[D] These are generically nonzero and first order in ε. Current mean, covariance and repeated third-cumulant entries do not determine this perturbation. Consequently a recursion based only on the current slice cannot be exact through first order for general admissible third cumulants. Its "cheap correction" must carry or reconstruct precisely the information absent from that slice.

[D] More generally, a linear encoding closed under every invertible transport U and able to return all diagonals must be injective on $\mathrm{Sym}^3(R^n)$. Otherwise a nonzero kernel tensor H would have H[v,v,v]=0 for every v, contradicting polarization. Such a universal linear state needs at least $n(n+1)(n+2)/6 = 179{,}481{,}600$ scalars at n=1024 [D].

[D] This does **not** rule out a nonlinear encoding, a restricted source family, a finite-horizon query representation, or an ensemble-specific approximation. Your existing factored source state is nonlinear in its legs. It is also not a proof that a given matrix-product cost is optimal.

## A common propagator does not provide a closed square update

[D] If an old propagator is P=VR, then its exact elementwise square is:

```
(VR)*(VR) = (V*V)(R*R)
 + 2 sum_(p<q) (V_:p*V_:q)(R_p:*R_q:)
```

[D] The off-diagonal pair terms are genuine additional coordinates. Keeping them introduces the symmetric-square space of dimension n(n+1)/2; dropping them is an approximation to the queried fluctuation, not an identity. A shared propagator or static hub side does not, on its own, remove their contraction cost. No new accuracy claim follows from that rewrite.

[Q/D] The paper describes factorization as preserving the selected diagram algorithm's output, with source-list growth producing quadratic dependence on depth. It does not supply a source-free, slice-only closure or an optimality theorem. See Wu et al., §4.3 and technical supplement §S.4.3. The identities above are independently derived.

# A cheap certificate for a future candidate

A necessary local test; passing it is not a whole-network accuracy result

[D] Your slice-to-slice failures already supply evidence against that approach. The following is a compact way to expose the missing target using stored legs; it is not a recommendation to retry a killed architecture. Use the captured L14→L15 pair and 64 fixed output indices I [D test setting]. Use the capture's actual $w_1$ and the next W.

```
V = diag(w1) W[:,I]                 # [n,q], q=64
D_ik = T_iik;  t_i = T_iii
E = D - diag(t)                     # zero diagonal
B = E V                            # [n,q]
F_repeated = (V*V).T @ (B+t[:,None]*V)
             + 2 (V*B).T @ V       # [q,q]
```

[D] This is the exact next-slice contribution of the tensor retaining only current repeated-index entries, on the selected output pairs. Independently compute the full query from the captured source blocks. For every Sym(X,Y,Z), form $x=V^TX$, $y=V^TY$, $z=V^TZ$ and add:

```
F_block = [(x*y) @ z.T + (x*z) @ y.T
                       + (y*z) @ x.T] / 3
F_full = sum_blocks F_block
H_query = F_full - F_repeated
```

[D] Project the source's native A,P and thin legs first, then reconstruct X,Y,Z in q-dimensional output space. There is no need to materialize full transported n×n legs. Use D,t reconstructed from the same source state, or verify that the captured D21,d3 agree with it, so a capture mismatch does not appear as hidden-sector error.

[D] Hold the newborn contribution fixed as well. Read its block from the next capture, restrict its readout to I, and call it F_new. Set F_ref=F_full+F_new. This normalization uses the total frozen next-layer slice, so a small old contribution is not judged against an artificially tight denominator.

## The gate with all other contributions held fixed

```
slice_error = ||offdiag(F_repeated+Gamma-F_full)||_F
              / ||offdiag(F_ref)||_F
diag_error  = ||diag(F_repeated+Gamma-F_full)||_2
              / ||diag(F_ref)||_2
```

[D] Both errors must be at most 0.02 [Q] on the sampled queries before a claim of preserving both channels is credible. A large $H_{query}$ rejects $\Gamma=0$; it does not reject every learned or state-dependent Γ. The numerator $||H_{query}-\Gamma||$ is the actual correction problem. A pass on this subset still requires all queried entries and every requested late layer, followed by the coupled mean-closure run.

## Arithmetic of this diagnostic

[D] With m sources, two native dense legs per source and the three displayed symmetric blocks, the dominant classical cost is:

```
4 m q n^2 + 18 m q^2 n
+ 2 n^2 q + 4 n q^2   FLOPs
+ thin-factor and elementwise work
```

[D] At n=1024, q=64 and m≤15 (allowing an extra block for the newborn), the displayed terms cost at most 2.473 u. Allowing up to 50 thin columns per source and conservative elementwise overhead keeps a straightforward implementation below about 3 u of arithmetic [D bound for those shapes]. Meter-specific overhead is unmeasured [E]. The diagnostic uses chain captures only; teacher statistics are reserved for checking the replacement's actual channel and mean accuracy.

**[S] Status:** no qualifying representation, 60%-cost implementation, or 2%-accurate replacement has been established here. The exact identities eliminate misleading shortcuts, but they do not establish that your requested architecture is impossible. I would require a proposed state to specify Γ and price its update before claiming a new direction.